

Oppgaver for group sessions uke 8 (16.02-20.02)

MNF130

Spring 2026

Oppgavene er hentet fra læreboken K. Rosen: Discrete Mathematics and its Applications, kapittel 1.6.

1. What rule of inference is used in each of these arguments?
 - (a) Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.
 - (b) Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.
 - (c) If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
 - (d) If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
 - (e) If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.
2. For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion for the premises.
 - (a) "If I play hockey, then I am sore the next day." "I use the whirlpool if I am sore." "I did not use the whirlpool."
 - (b) "If I work, it is either sunny or partly sunny." "I worked last Monday or I worked last Friday." "It was not sunny on Tuesday." "It was not partly sunny on Friday."
 - (c) "All insects have six legs." "Dragonflies are insects." "Spiders do not have six legs." "Spiders eat dragonflies."
 - (d) "Every student has an Internet account." "Homer does not have an Internet account." "Maggie has an Internet account."
 - (e) "All foods that are healthy to eat do not taste good." "Tofu is healthy to eat." "You only eat what tastes good." "You do not eat tofu." "Cheeseburgers are not healthy to eat."

- (f) “I am either dreaming or hallucinating.” “I am not dreaming.” “If I am hallucinating, I see elephants running down the road.”
3. For each of these arguments determine whether the argument is correct or incorrect and explain why.
- (a) All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.
 - (b) Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.
 - (c) All parrots like fruit. My pet bird is not a parrot. Therefore, my pet does not like fruit.
 - (d) Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.
4. What is wrong with this argument? Let $H(x)$ be “ x is happy”. Given the premise $\exists x H(x)$, we conclude that $H(\text{Lola})$. Therefore, Lola is happy.
5. Determine whether each of these arguments is valid. If an argument is correct, what rule of inference is being used? If it is not, what logical error occurs?
- (a) If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$. Then $n > 1$.
 - (b) If n is a real number with $n > 3$, then $n^2 > 9$. Suppose that $n^2 \leq 9$. Then $n \leq 3$.
 - (c) If n is a real number with $n > 2$, then $n^2 > 4$. Suppose that $n \leq 2$. Then $n^2 \leq 4$.
6. Identify the error or errors in this argument that supposedly shows that if $\exists x P(x) \wedge \exists x Q(x)$ is true then $\exists x (P(x) \wedge Q(x))$ is true.
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| 1. | $\exists x P(x) \wedge \exists x Q(x)$ | Premise |
| 2. | $\exists x P(x)$ | Simplification from (1) |
| 3. | $P(c)$ | Existential instantiation from (2) |
| 4. | $\exists x Q(x)$ | Simplification from (1) |
| 5. | $Q(c)$ | Existential instantiation from (4) |
| 6. | $P(c) \wedge Q(c)$ | Conjunction from (3) and (5) |
| 7. | $\exists x (P(x) \wedge Q(x))$ | Existential generalization from (6) |
7. Identify the error or errors in this argument that supposedly shows that if $\forall x (P(x) \vee Q(x))$ is true then $\forall x P(x) \vee \forall x Q(x)$ is true.

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| 1. | $\forall x(P(x) \vee Q(x))$ | Premise |
| 2. | $P(c) \vee Q(c)$ | Universal instantiation from (1) |
| 3. | $P(c)$ | Simplification from (2) |
| 4. | $\forall xP(x)$ | Universal generalization from (3) |
| 5. | $Q(c)$ | Simplification from (2) |
| 6. | $\forall xQ(x)$ | Universal generalization from (5) |
| 7. | $\forall xP(x) \vee \forall xQ(x)$ | Conjunction from (4) and (6) |